

TRACE TEST — SHOULD AI MAKE THIS DECISION?

Scenario Analysed: AI in Loan Approval

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1. Scenario Description

Banks and digital lenders use AI models to decide whether to approve a loan and at what interest rate, by analysing income, credit score and repayment history to predict default risk. This speeds up lending and can extend credit to people with thin credit files, but also risks hidden bias, since the model learns from historical data that may reflect past discrimination.

2. Applying the TRACE Test

Element	Analysis
T – Task	AI can automate the data-driven, repetitive part : pulling credit data, calculating a default-risk score, checking eligibility rules, and fast-tracking clearly strong or weak applications.
R – Risk	Main harms: wrongful denial and unequal treatment of vulnerable groups. A US fair-lending audit of fintech lender Upstart found approval disparities for Black applicants despite no direct race proxy, and studies link algorithmic bias to historical redlining in training data. Minorities, low-income and thin-credit-file applicants are most exposed.
A – Authority	A qualified loan/credit officer must be able to review, override or halt an AI recommendation — especially for borderline or rejected cases — not just rubber-stamp it. Human judgment catches context the model cannot see, e.g. a temporary income gap due to illness.
C – Contestability	Applicants must get a plain-language reason for rejection, a route to human review, and a way to correct wrong data. US fair-lending law (ECOA, Fair Housing Act) already mandates reasons for adverse credit decisions — without this, an automated rejection cannot be challenged.
E – Evidence	Mixed. Lenders claim AI is more consistent and widens credit access, but independent audits (e.g. Upstart monitorship) show real disparities, and research confirms historical bias persists unless corrected. No strong proof yet that AI is fairer than supervised human underwriting — only that it is more scalable.

3. What AI Should Assist With vs. What Must Remain Human

AI may assist with	Must remain a human decision
- Collecting and verifying financial data - Calculating a risk/default score - Fast-tracking clearly eligible, low-risk applications - Flagging inconsistent or suspicious documents	- Final rejection of a borderline or ambiguous application - Handling appeals and hardship or special-circumstance cases - Deciding whether a flagged bias pattern requires policy change - Any decision that cannot be fully explained to the applicant

4. Non-Negotiable Ethical Safeguard

Mandatory human review and a right to appeal every rejected application. No applicant should be finally denied credit purely by an algorithm, with no route to a human who can check for errors or bias.

5. Conclusion

Loan approval should be **partially automated, not fully automated**. AI suits the data-processing and risk-scoring stage, improving speed and consistency, but persistent evidence of bias and the seriousness of the harm mean final approval/rejection — especially borderline cases — must stay human-led, with transparent reasons and a working appeal process.

References

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